

Answer Key — Biology Grade 11

Sub-Strand 3.2: Growth and Development in Animals • Lesson 2: Stepwise Growth: Why Hard-Shelled Animals Grow in Jumps

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	A	C	B	D	D	A

USE AFTER: PREDICT PHASE — Predict: sketching a graph of a nymph's body length against time over 10 weeks

Q1. A grasshopper nymph in a school garden in Kisumu is hatched and its body length is measured every few days for 10 weeks. Its body is covered by a hard exoskeleton. What shape should the graph of length against time have?

- A. A smooth curve that rises steadily, like a child's height
- B. A staircase shape with flat parts (plateaus) and sudden rises**
- C. A flat line, because insects do not grow after hatching
- D. A straight line sloping upwards from the first day

Correct answer: B. A rigid exoskeleton cannot stretch, so length stays the same for a period and then jumps after moulting, giving plateaus and sudden rises. A smooth curve is the common misconception about growth.

USE AFTER: OBSERVE PHASE — Observe: plotting length against time from the insect growth dataset and marking moulting days

Q2. A locust nymph's length stayed at 14 mm from Day 12 to Day 17. On Day 18 it measured 19 mm. By how many millimetres did its length increase between Day 17 and Day 18?

- A. 5 mm**
- B. 19 mm
- C. 14 mm
- D. 33 mm

Correct answer: A. Increase = new length - old length = 19 - 14 = 5 mm. This sudden jump marks the day the nymph moulted.

USE AFTER: OBSERVE PHASE — Observe: plotting mass for comparison with the length graph

Q3. Learners plot both length and mass of a desert locust nymph over 35 days. Which statement best describes what they are likely to see between moults?

- A. Length and mass both stay flat the whole time until the adult stage
- B. Length rises smoothly while mass stays flat, then mass jumps at a moult
- C. Mass can keep increasing gradually while length stays flat, then length jumps at a moult**
- D. Both length and mass drop each time the nymph feeds

Correct answer: C. The nymph keeps eating, so mass can rise gradually between moults. The hard exoskeleton holds the length constant until the nymph moults, so length rises in jumps.

USE AFTER: EXPLAIN PHASE — Explain: CER (Claim-Evidence-Reasoning) on why the nymph's length increases in steps

Q4. A grasshopper nymph keeps eating sukuma wiki, but its length does not change from Day 12 to Day 17. What is the best scientific reason?

- A. It stops digesting food during those days
- B. Its hard exoskeleton cannot stretch, so its length cannot increase until it moults**
- C. The nymph shrinks slightly and then regrows
- D. The old skin is thin and peels off slowly like sunburned skin

Correct answer: B. The rigid exoskeleton limits body size. Length can only increase after ecdysis, so the graph is flat during the intermoult period even though the insect feeds.

USE AFTER: EXPLAIN PHASE — Explain: teacher addressing the misconception that insects peel like sunburned skin

Q5. A learner finds an empty exuvia on a maize leaf and says, "It is just a thin peel of skin, like sunburned skin." Which statement corrects this idea?

- A. The exuvia is the new skin which has not hardened yet
- B. The exuvia is a dead young insect that did not survive
- C. The exuvia is a thin peel that comes off only from the surface of the body
- D. The exuvia is a complete hard outer casing, shaped like the insect with legs, left behind when the insect moults**

Correct answer: D. The exuvia is the whole old cuticle (exoskeleton), split along the back and shed during ecdysis. It is not a thin peel.

USE AFTER: EXPLAIN PHASE — Explain: CER reasoning on why the grasshopper is bigger than its shell, and the Gallery Walk

Q6. A newly emerged grasshopper is larger than the exuvia it has just left behind. Which explanation is correct?

- A. The new cuticle hardened first and then the insect grew inside it
- B. The old exuvia shrank after it was shed
- C. The old exoskeleton stretched slowly as the grasshopper grew inside it
- D. After splitting the old cuticle, the insect swells and expands while the new cuticle is still soft, before it hardens**

Correct answer: D. Growth happens just after moulting. The insect expands while the new cuticle is soft, and then the cuticle hardens at the larger size, so the insect ends up bigger than the old exuvia.

USE AFTER: MODEL BUILDING PHASE — Model: revised labelled diagram sequence of nymph moulting, linked to the stepwise graph

Q7. A grasshopper nymph is somehow unable to moult. Based on the lesson, what would happen to its body length?

- A. It would stay limited by the hard exoskeleton and could not increase**
- B. It would increase in steps without needing a new cuticle
- C. It would grow only in mass and shrink in length
- D. It would grow smoothly because the exoskeleton would stretch

Correct answer: A. Body length can increase only after ecdysis, when the new cuticle is soft. Without moulting, the rigid exoskeleton prevents any increase in size.